Problem set 2 – Econ 5030

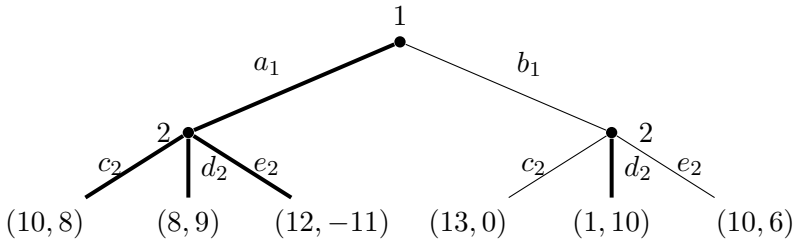
1. Iterated deletion

A) R1: n_G str. dom. by l_G ($3 > 0, 1, 1$); n_P str. dom. by l_P ($2 > 0$, $3 > 1$, $3 > 1$). (h_P vs l_P only weak: tie $2 = 2$ at n_G.) R2: h_G str. dom. l_G ($4 > 3$, $5 > 3$); h_P str. dom. l_P ($4 > 3$, $5 > 3$). Solution set: $\{(h_G, h_P)\}$, payoffs $(5, 5)$.

B) BR_G: $n_P \to l_G$, $l_P \to h_G$, $h_P \to h_G$. BR_P: $n_G \to \{l_P, h_P\}$, $l_G \to h_P$, $h_G \to h_P$. NE $= \{(h_G, h_P)\}$ (unique, also in mixed, since dominance solvable).

2. Normal form

1. Tree, payoffs (u_1, u_2); thick = backward induction.



2. $S_1 = \{a_1, b_1\}$. $S_2 = \{xy : x, y \in \{c_2, d_2, e_2\}\}$, x after a_1, y after b_1 (9 strategies).

3.

	cc	cd	ce	dc	dd	de	ec	ed	ee
a_1	$10, 8$	$10, 8$	$10, 8$	$8, 9$	$8, 9$	$8, 9$	$12, -11$	$12, -11$	$12, -11$
b_1	$13, 0$	$1, 10$	$10, 6$	$13, 0$	$1, 10$	$10, 6$	$13, 0$	$1, 10$	$10, 6$

4. Vs a_1: P2 needs d_2 after a_1 ($9 > 8 > -11$); then a_1 (8) $\geq b_1$ only if $y = d_2$ ($13, 1, 10$). Vs b_1: P2 needs d_2 after b_1; then b_1 gives $1 < 10, 8, 12$. Unique pure NE: (a_1, d_2d_2), payoffs $(8, 9)$.

5. No. Backward induction: d_2 after a_1, d_2 after b_1 ($10 > 6 > 0$); P1: $8 > 1 \Rightarrow a_1$. SPE $= (a_1, d_2d_2) =$ the NE. The punishment d_2 after b_1 is P2's best reply there, so the threat is credible.

3. Timing and strategies

2.1 No info: strategy = action. $S_1 = S_2 = \{0, 1\}$; 2×2 matching pennies, no pure NE, mixed $(\frac{1}{2}, \frac{1}{2})$.

2.2 $S_1 = \{0, 1\}$. P2 conditions on P1's move: $S_2 = \{(0, 0), (0, 1), (1, 0), (1, 1)\}$ (x after 0, y after 1) = always 0, copy, opposite, always 1.

2.3 Same actions, different strategy sets: 2×2 vs 2×4.

	$(0, 0)$	$(0, 1)$	$(1, 0)$	$(1, 1)$
0	$1, 0$	$1, 0$	$0, 1$	$0, 1$
1	$0, 1$	$1, 0$	$0, 1$	$1, 0$

Timing/info enter only through strategy sets. Sequential: P2's "opposite" $(1, 0)$ always wins; NE $= (0, (1, 0)), (1, (1, 0))$, payoffs $(0, 1)$. Simultaneous: each wins w.p. $\frac{1}{2}$.

4. Third party observation

2.4 No. With payoff = own \$, (M, M) is not NE: vs M, deviate to L (\$30 > \$18) if first \$ is the row player's, or to H (\$20 > \$18) if first \$ is P1's. This rejects "maximize own money + NE", not game theory: payoffs are utilities over outcomes, and the experimenter controls money, not preferences.

2.5 $G = (N, \{S_i\}, \{u_i\})$: players; strategies = complete contingent plans (an action at every info set, even unreached); $u_i : \prod_j S_j \to \mathbb{R}$ represents i's preferences over induced outcomes (ordinal for pure, vNM for mixed). Players choose strategies simultaneously and independently; timing/info appear only through S_i. Here $S_i = \{L_i, M_i, H_i\}$ and u_i need not equal own \$.

2.6 Fairness/social preferences: $u_i = x_1 + x_2 + \min\{x_1, x_2\}$ (same for both players).

	L_1	M_1	H_1
L_2	$24, 24$	$50, 50$	$17, 17$
M_2	$50, 50$	$54, 54$	$28, 28$
H_2	$17, 17$	$28, 28$	$15, 15$

M strictly dominant ($50 > 24, 17$; $54 > 50, 28$; $28 > 17, 15$) $\Rightarrow$ unique NE (M_1, M_2).

Rationality test? No, not alone. Utilities are unobserved and any profile can be made the unique NE by choice of preferences (as in 2.6). The data only test rationality jointly with an assumed utility (e.g. own money). Need cross-game consistency (revealed preference) or elicited preferences/beliefs.

5. SPNE (P2 observes P1, then picks U_2/D_2.)

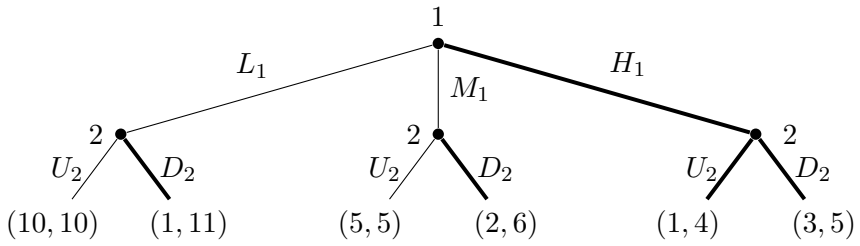
1. $S_1 = \{L_1, M_1, H_1\}$; $S_2 = \{U, D\}^3$, $x_L x_M x_H =$ action after L_1, M_1, H_1 (8 strategies); $u_i(k_1, x_L x_M x_H) = u_i(k_1, x_k)$.

2.

	UUU	UUD	UDU	UDD	DUU	DUD	DDU	DDD
L_1	$10, 10$	$10, 10$	$10, 10$	$10, 10$	$1, 11$	$1, 11$	$1, 11$	$1, 11$
M_1	$5, 5$	$5, 5$	$2, 6$	$2, 6$	$5, 5$	$5, 5$	$2, 6$	$2, 6$
H_1	$1, 4$	$3, 5$	$1, 4$	$3, 5$	$1, 4$	$3, 5$	$1, 4$	$3, 5$

P2 plays D on path ($11 > 10$, $6 > 5$, $5 > 4$). L_1: gets $1 <$ M_1's ≥ 2, no NE. M_1: need $x_L = D$, $x_H = U$. H_1: need $x_L = x_M = D$. NE: (M_1, DDU) $(2, 6)$ and (H_1, DDD) $(3, 5)$.

3.



P2: D_2 at every node. P1: $1, 2, 3 \Rightarrow H_1$. SPE: (H_1, DDD), payoffs $(3, 5)$. (M_1, DDU) is not SPE: it uses the incredible threat U_2 after H_1 $(4 < 5)$.